

JAY DARJI

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Machine Learning Engineer with **2+ years** building **deep learning** and **NLP** systems across **Python**, **PyTorch**, and **LLMs**. Delivered model pipelines end-to-end, from research to server-side deployment on AWS; published researcher with **80+ citations** across 3 papers; MS in EE (AI); available **immediately**.

TECHNICAL SKILLS

Languages: Python, Java, SQL (MySQL), R

AI and ML: Deep Learning, NLP, PyTorch, TensorFlow, Scikit-learn, Computer Vision, CNN-LSTM

LLM and MLOps: LLMs, RAG, SBERT, Flask REST APIs, Docker, AWS (EC2, S3), CI/CD

Backend and Data: Spring Boot, RESTful APIs, ETL Pipelines, Git, Agile

EDUCATION

San Jose State University, San Jose

GPA: 3.71/4

Master of Science in Electrical Engineering (Concentration: Artificial Intelligence)

Aug 2024 – May 2026

Relevant Courses: Machine Learning, Neural Networks, Reinforcement Learning, Computer Vision, AI for DSP

Institute of Infrastructure Technology Research and Management, Ahmedabad

GPA: 3.2/4

Bachelor of Technology in Electrical Engineering

Aug 2017 – May 2021

EXPERIENCE

Machine Learning Research Assistant

Jun 2025 – Feb 2026

San Jose State University, San Jose

- Engineered a **server-side VLM pipeline** (DINOv3 ViT + GPT-4o-mini), reaching **83.25% accuracy**.
- Architected a **multi-agent framework** automating feature extraction and model training.
- Implemented automated **data wrangling** workflows with auditable model performance tracking.
- Delivered complex analytical results as clear reports for non-technical stakeholders.

Data Scientist

Sep 2022 – May 2024

Rhenix Lifesciences LLP, Hyderabad

- Built a **CNN-LSTM** model on clinical **database** records; achieved **10 mg/dL RMSE** at 15-min horizon.
- Engineered a **Random Forest** classifier with cross-validation; attained **93% accuracy**, 0.96 AUC.
- Designed **comparative experiments** at 15- and 60-min intervals, outperforming published benchmarks.
- Constructed **ETL pipelines** with data quality validation; built **Tableau** dashboards.

Back-End Software Developer

Dec 2021 – Apr 2022

Cognizant Technology Solutions Pvt. Ltd., Bengaluru

- Built **Java Spring Boot** microservices, maintaining version-controlled code in **Agile Git** workflows.
- Designed and integrated **RESTful APIs** to support client-server data exchange.
- Contributed to **SQL** query optimization, helping reduce average **database** execution time by **20%**.
- Maintained technical documentation supporting team-wide **server-side** process improvement.

Machine Learning Research Intern

Jun 2019 – Aug 2019

Institute of Infrastructure Technology Research and Management, Ahmedabad

- Validated an ensemble classifier through experimental design; attained **95% accuracy**.
- Co-authored a **peer-reviewed paper**; mentored 10 students on **ML** experimentation and research.

ACADEMIC PROJECTS

Memory-Constrained LLM Inference Optimization on Commodity GPUs

- Benchmarked **FlashAttention-2** and SDPA kernels on a 4GB-VRAM GPU; raised max context **18.9%**.
- Applied 4-bit **NF4** and **KV-cache INT4 quantization** across six LLMs via bitsandbytes.
- Evaluated zero-shot accuracy (ARC-Easy, BoolQ, HellaSwag) within **2%** of full-precision baseline.

NLP Summarization Service: Back-End Deployment on AWS with CI/CD Pipeline

- Built a **RAG-based NLP** service with data transformation and automated retraining; ROUGE-1: **0.51**.
- Deployed on **AWS EC2** via **Docker**; automated **CI/CD** with GitHub Actions; monitored pipeline health.
- Exposed a **deep learning** model via **Flask RESTful API**; documented data assumptions and failure modes.

AI-Powered Resume Matching Service with Embedding Models and LLM Inference

- Built a **server-side NLP** matching engine using SBERT, OpenAI, and Gemini embeddings.
- Fine-tuned SBERT on 150 labeled pairs; boosted match accuracy from **61% to 74%**.
- Delivered **LLM-generated** fit summaries via Streamlit UI; **API** response averaged under 5 seconds.

RESEARCH PUBLICATIONS

- Sleep disorders identification via wavelet-based EOG and EMG features DOI: 10.1016/j.complbiomed.2022.105224
- Handling missing data in time-series data from wearables DOI: 10.5772/intechopen.1002536
- Efficient use of binned data for imputing uni-variate time-series data DOI: 10.3389/fdata.2024.1422650